Finite Trigonometry over $\mathbb{F}_7$

Angles in $\mathbb{Z}/8$, the Conic $x^2 + y^2 = 1$, and a First Look at Harmonic Analysis

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Abstract

Over $\mathbb{F}_7$ the conic $x^2 + y^2 = 1$ has exactly eight points. They form the norm-one group of $\mathbb{F}_{49} = \mathbb{F}_7[\mathrm{i}]/(\mathrm{i}^2 + 1)$, a cyclic group of order eight. Writing $\alpha = 2 + 2\mathrm{i}$ and $\alpha^k = \cos k + \mathrm{i} \sin k$ defines a finite trigonometry with angle group $\mathbb{Z}/8$ and values in $\mathbb{F}_7$. We prove the Pythagorean, addition, doubling and periodicity identities, draw the finite unit circle and the angle group, give the stereographic (half-angle) coordinate, and draw the discrete waveforms, and compare the finite angle group with the Euclidean angle axioms. A final section introduces the harmonic layer: the characters of $\mathbb{Z}/8$, the finite Fourier transform, and the difference between the $\mathbb{F}_7$-valued trigonometric functions and the $\mathbb{C}$-valued characters. We also explain why the angle measure is discrete and why half-angles need not exist. This is an algebraic and finite theory; it is not a limit-based real analysis, and the note says explicitly what is and is not claimed.

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1 Introduction

Over the real numbers the functions $\cos\theta$ and $\sin\theta$ are defined by the unit circle $x^2 + y^2 = 1$ and by angles in $\mathbb{R}/2\pi\mathbb{Z}$. Over a finite field the circle is a finite set, the angle group is finite, and the values are field elements. The algebraic identities survive; the limiting and continuity notions do not.

For $\mathbb{F}_7$ the situation is particularly clean. The conic

$$C: \quad x^2 + y^2 = 1$$

has exactly $7+1 = 8$ points, and these points form a cyclic group of order 8 under the multiplication of the field $\mathbb{F}_{49}$. This gives a finite analogue of the circle group, with angle group $\mathbb{Z}/8$, and a finite trigonometry whose identities are polynomial identities over $\mathbb{F}_7$.

The plan is:

1. define the field $\mathbb{F}_{49}$ and the norm-one group, and ask whether the norm induces a metric or a topology (Section 2);
2. define $\cos k$ and $\sin k$, and prove the basic identities (Sections 3–4);
3. draw the finite unit circle and the angle group (Sections 5–6);
4. compare the finite angle group with the Euclidean angle axioms (Section 7);
5. give the stereographic/half-angle coordinate (Section 8);
6. draw the discrete waveforms (Section 9);
7. introduce the harmonic layer: characters of $\mathbb{Z}/8$ and the finite Fourier transform (Section 10);
8. state what is not claimed and what comes next (Section 11).

Throughout, $\mathbb{F}_7 = \{0, 1, \ldots, 6\}$ with arithmetic modulo 7; standard background on finite fields is in [3]. The element i is a formal square root of -1; since -1 is not a square in $\mathbb{F}_7$ (the squares are $0, 1, 2, 4$), the ring $\mathbb{F}_7[\mathrm{i}]/(\mathrm{i}^2 + 1)$ is a field with 49 elements.

2 The field $\mathbb{F}_{49}$ and the norm-one group

Definition 2.1 (Norm and norm-one group)**.** For $z = x + y\mathrm{i} \in \mathbb{F}_{49}$ define the norm

$$N(z) = z\bar{z} = (x + y\mathrm{i})(x - y\mathrm{i}) = x^2 + y^2 \in \mathbb{F}_7.$$

Let $G = \{z \in \mathbb{F}_{49}^{\times} : N(z) = 1\}$.

Proposition 2.2. *G is a cyclic group of order 8.*

Proof. The norm map $N : \mathbb{F}_{49}^{\times} \to \mathbb{F}_7^{\times}$ is a surjective homomorphism of finite groups (the norm of a generator of $\mathbb{F}_{49}^{\times}$ generates $\mathbb{F}_7^{\times}$). Hence $|G| = 48/6 = 8$. A subgroup of the cyclic group $\mathbb{F}_{49}^{\times}$ is cyclic, so G is cyclic. □

Proposition 2.3. *Let $\alpha = 2 + 2\mathrm{i}$. Then α has order 8, and $G = \langle\alpha\rangle$.*

Proof. We compute

$$\alpha^2 = 4 + 8\mathrm{i} + 4\mathrm{i}^2 = 8\mathrm{i} = \mathrm{i}, \qquad \alpha^4 = \mathrm{i}^2 = -1, \qquad \alpha^8 = 1.$$

Since $\alpha^2 = \mathrm{i} \neq 1$ and $\alpha^4 = -1 \neq 1$, the order of α is 8. It lies in G because $N(\alpha) = 2^2 + 2^2 = 8 = 1$, and G has order 8; therefore $G = \langle\alpha\rangle$. □

The eight powers are

k	0	1	2	3	4	5	6	7
α^k	1	$2+2\mathrm{i}$	i	$5+2\mathrm{i}$	-1	$-2-2\mathrm{i}$	$-\mathrm{i}$	$2-2\mathrm{i}$

and their coordinates (x, y) are the eight points

$$(1,0),\ (2,2),\ (0,1),\ (5,2),\ (6,0),\ (5,5),\ (0,6),\ (2,5).$$

They are exactly the solutions of $x^2 + y^2 = 1$ in $\mathbb{F}_7^2$.

2.1 Does the norm induce a metric or a topology?

The norm takes values in $\mathbb{F}_7$, not in an ordered set. A metric in the analytic sense is a function d with values in an ordered set, positive on distinct points and satisfying the triangle inequality. The candidate $d(z, w) = N(z - w)$ fails at the first step: $\mathbb{F}_7$ has no canonical order, so "$N(z - w) \geq 0$" and "$N(z - w) < r$" are not meaningful. The norm is anisotropic ($N(z) = 0$ iff $z = 0$, because -1 is a nonsquare), but anisotropy is nondegeneracy, not a metric.

The only absolute value on $\mathbb{F}_7$ is the trivial one ($|0| = 0$, $|x| = 1$ for $x \neq 0$). Composing it with the norm gives $|N(z - w)| \in \{0, 1\}$, the discrete metric; the induced topology is the discrete topology. The Zariski topology on the finite set of $\mathbb{F}_7$-points is also discrete. In this analytic sense the norm gives no interesting metric or topology.

In the algebraic sense the answer is different and richer. The norm is a nondegenerate anisotropic quadratic form; its polar form is the dot product $u \cdot v$, and its isometry group is the orthogonal group $\mathrm{O}_2(\mathbb{F}_7)$. The rotation subgroup is exactly the norm-one group $G = \mathrm{SO}_2(\mathbb{F}_7) \cong \mathbb{Z}/8$, and the conic C is the unit sphere. So the norm does induce a *metric vector space* (a quadratic space) with orthogonality, isometries and a unit sphere; it does not induce a metric space in the topological sense.

What the norm does induce geometrically is a stratification. The fibres are

$$N^{-1}(0) = \{0\}, \qquad N^{-1}(a) \cong C \quad (a \in \mathbb{F}_7^\times),$$

each nonzero fibre being an 8-point conic and a G-torsor under $z \mapsto \alpha^k z$. The six nonzero fibres partition $\mathbb{F}_{49}^\times$; Figure 1 shows them. This is the finite analogue of polar coordinates: radius $a \in \mathbb{F}_7^\times$, angle $k \in \mathbb{Z}/8$. The decomposition is a torsor decomposition, not a canonical group product: the exact sequence $1 \to \mathbb{Z}/8 \to \mathbb{F}_{49}^\times \to \mathbb{F}_7^\times \to 1$ does not split as a direct product (the two orders have gcd 2), and choosing a base point in each fibre is a noncanonical section.

The cyclic word metric on $G = \mathbb{Z}/8$ (the one used for the angle group) is not induced by the norm: the norm is constant 1 on G. It is induced by the cyclic order of the angle group.

To obtain a genuine metric and topology from the norm one must lift. The unramified extension $\mathbb{Q}_{49}/\mathbb{Q}_7$ has residue field $\mathbb{F}_{49}$; the 7-adic absolute value gives a metric on $\mathbb{Q}_{49}$, the field norm $N_{\mathbb{Q}_{49}/\mathbb{Q}_7}$ is continuous, and the finite norm is its reduction modulo 7. This is the natural bridge from the finite norm to p-adic analysis; the finite field itself only carries the discrete topology.

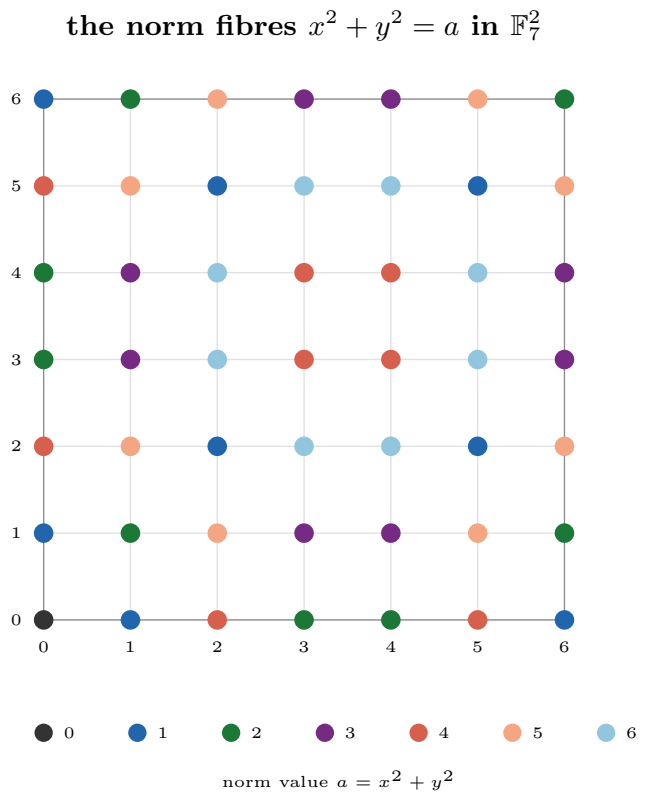


Figure 1: The norm fibres $x^2+y^2 = a$ in $\mathbb{F}_7^2$: the origin ($a = 0$) and six 8-point conics ($a = 1, \ldots, 6$). The colours are the norm value; the legend is at the bottom. This is a stratification, not a metric topology.

question	answer
distance $d(z,w) = N(z-w)$ in an ordered set?	no: $\mathbb{F}_7$ is unordered; only the trivial absolute value; discrete topology
metric vector space?	yes: nondegenerate anisotropic quadratic form; dot product; $\mathrm{O}_2(\mathbb{F}_7)$
topology from the norm?	discrete; the Zariski topology on the finite points is discrete
fibres?	$\{0\}$ and six 8-point conics; G-torsors
genuine metric or topology?	lift to $\mathbb{Q}_{49}/\mathbb{Q}_7$ or $W(\mathbb{F}_{49})$; 7-adic absolute value
metric on $\mathbb{Z}/8$?	cyclic word metric, from the group, not from N

Table 1: What the norm does and does not induce. The analytic notions (distance, order, continuity) are absent; the algebraic notions (quadratic form, isometries, torsors) are present; the analytic version requires a lift.

3 Cos and sin

Definition 3.1 (Finite cosine and sine)**.** For $k \in \mathbb{Z}/8$, write

$$\alpha^k = \cos k + \mathrm{i} \sin k, \qquad \cos k,\ \sin k \in \mathbb{F}_7.$$

The values are given in Table 2.

k	α^k	$\cos k$	$\sin k$
0	1	1	0
1	$2+2\mathrm{i}$	2	2
2	i	0	1
3	$5+2\mathrm{i}$	5	2
4	-1	6	0
5	$-2-2\mathrm{i}$	5	5
6	$-\mathrm{i}$	0	6
7	$2-2\mathrm{i}$	2	5

Table 2: The eight angles and their cosines and sines in $\mathbb{F}_7$.

Proposition 3.2 (Basic identities)**.** *For all $k, l \in \mathbb{Z}/8$:*

1. *(periodicity)* $\cos(k+8) = \cos k$ *and* $\sin(k+8) = \sin k$*;*
2. *(parity)* $\cos(8-k) = \cos k$ *and* $\sin(8-k) = -\sin k$*;*
3. *(Pythagoras)* $\cos^2 k + \sin^2 k = 1$.

Proof. (1) is $\alpha^{k+8} = \alpha^k$. (2) is $\alpha^{-k} = \overline{\alpha^k}$, using $\bar{\alpha} = \alpha^{-1}$. (3) is $N(\alpha^k) = 1$. □

Remark 3.3. The values are in $\mathbb{F}_7$ and the angle is in $\mathbb{Z}/8$; there is no continuous graph and no real-valued sine wave. The identities are polynomial identities in $\mathbb{F}_7$, and they are verified for all k by the script `verify.py`.

3.1 The angle measure is discrete; there are no radians

The protractor is the isomorphism

$$\mathbb{Z}/8 \longrightarrow G, \qquad k \longmapsto \alpha^k.$$

The angle is the group element; the label k is its coordinate. This is the whole of the finite angle measure. There is no real-valued additive measure, no arc length, and no radian: $\mathbb{F}_7$ is unordered, the conic is a finite set, and there is no continuous arc to length.

One can put a metric on $\mathbb{Z}/8$, the cyclic word metric

$$\ell(k) = \min(k, 8-k),$$

but it is a size, not a measure: it is not additive. For example, $\ell(3+4) = \ell(7) = 1$ while $\ell(3) + \ell(4) = 3 + 4 = 7$. The only additive "measure" is the group element itself; the word metric is the metric of the Cayley graph, not a protractor.

The unit is the choice of generator: α generates G, and the automorphisms of $\mathbb{Z}/8$ are multiplication by $1, 3, 5, 7$. Changing the generator relabels the eight angles but does not change the group. The full turn is $8 = 0$, the straight angle is 4, and the right angle is 2; calling the generator a "degree" is a labelling choice, not a radian. The periodic waveforms of Figure 11 show the period 8; there is no interpolation between the eight points.

4 Addition and doubling

Proposition 4.1 (Addition formulas)**.** *For all $k, l \in \mathbb{Z}/8$,*

$$\cos(k+l) = \cos k \cos l - \sin k \sin l, \qquad \sin(k+l) = \sin k \cos l + \cos k \sin l.$$

Proof. Multiply:

$$\alpha^{k+l} = (\cos k + \mathrm{i} \sin k)(\cos l + \mathrm{i} \sin l) = (\cos k \cos l - \sin k \sin l) + \mathrm{i}(\sin k \cos l + \cos k \sin l).$$

The real and imaginary parts are the two displayed identities. □

Corollary 4.2 (Doubling)**.** *For all $k \in \mathbb{Z}/8$,*

$$\cos 2k = \cos^2 k - \sin^2 k, \qquad \sin 2k = 2 \sin k \cos k.$$

The addition formula is the group law of $\mathbb{Z}/8$ transported to the conic: the map

$$k \longmapsto (\cos k, \sin k) = \alpha^k$$

is an isomorphism $\mathbb{Z}/8 \to G = C(\mathbb{F}_7)$. The doubling map $k \mapsto 2k$ is an endomorphism of $\mathbb{Z}/8$ with kernel $\{0, 4\}$; it is two-to-one, just like the real doubling map on the circle. It is *not* an automorphism (the automorphisms of $\mathbb{Z}/8$ are multiplication by $1, 3, 5, 7$).

Example 4.3. For $k = 1$: $\cos 1 = 2$, $\sin 1 = 2$, so $\cos 2 = 4 - 4 = 0$ and $\sin 2 = 2 \cdot 2 \cdot 2 = 8 = 1$, which is the table entry for $k = 2$.

4.1 Half-angles and division by two

The doubling formula gives a map $\mathbb{Z}/8 \to \mathbb{Z}/8$, $k \mapsto 2k$. Its kernel is $\{0, 4\}$ and it is two-to-one (Figure 8). The half-angle equation is

$$2h = k.$$

It is solvable if and only if k is even; when solvable it has two solutions, h and $h + 4$. For example, the right angle $k = 2$ has $h = 1, 5$, the straight angle $k = 4$ has $h = 2, 6$, and the odd angles $k = 1, 3, 5, 7$ have no half-angle at all.

The point is the difference between the field and the angle group:

- in the field $\mathbb{F}_7$, 2 is invertible ($2^{-1} = 4$), so $2x = a$ has a unique solution for every a;
- in the angle group $\mathbb{Z}/8$, 2 is not invertible, so $2h = k$ may have no solution, and when it has one it has two.

This is why the half-angle formula for the *square* of the cosine is well-defined even when the half-angle is not:

$$\cos^2 h = \frac{1 + \cos k}{2} \qquad (2h = k),$$

because the right-hand side only uses the field inverse of 2. The two solutions h and $h + 4$ give $(\cos h, \sin h)$ and $(-\cos h, -\sin h)$; the half-angle is defined only up to sign.

Table 3 summarizes the two equations. The consequences are:

- there is no global half-angle function $\mathbb{Z}/8 \to \mathbb{Z}/8$;
- there is no global angle bisector (the bisector exists only for even k, and then it is a line with two orientations);
- the inscribed-angle theorem has only a two-to-one shadow (Section 7);
- the stereographic coordinate t (the slope from P_0, the finite analogue of $\tan(k/2)$) is a rational coordinate on the seven angles $k \neq 4$, not a group homomorphism.

In Euclidean geometry the angle group $\mathbb{R}/2\pi\mathbb{Z}$ is divisible: $2h = k$ always has two solutions, h and $h + \pi$. The finite group $\mathbb{Z}/8$ is not two-divisible, and the failure is not a technicality: it is the absence of division in $\mathbb{Z}/8$.

equation	in the field $\mathbb{F}_7$	in the angle group $\mathbb{Z}/8$
$2x = a$	unique solution $x = a/2$	2 if a even; 0 if a odd
half-angle $2h = k$	unique $h = k/2$	2 if k even; 0 if k odd

Table 3: The field inverse of 2 and the angle equation $2h = k$. The field has $2^{-1} = 4$; the group $\mathbb{Z}/8$ has no division by 2.

5 The finite unit circle

The conic C is isomorphic to the projective line $\mathbb{P}^1(\mathbb{F}_7)$; the eight points of $C(\mathbb{F}_7)$ are the eight points of $\mathbb{P}^1(\mathbb{F}_7)$. The angle k is one coordinate on this set; the stereographic coordinate t of Section 8 is another. Neither coordinate is more intrinsic than the other: the intrinsic object is the conic together with its group law.

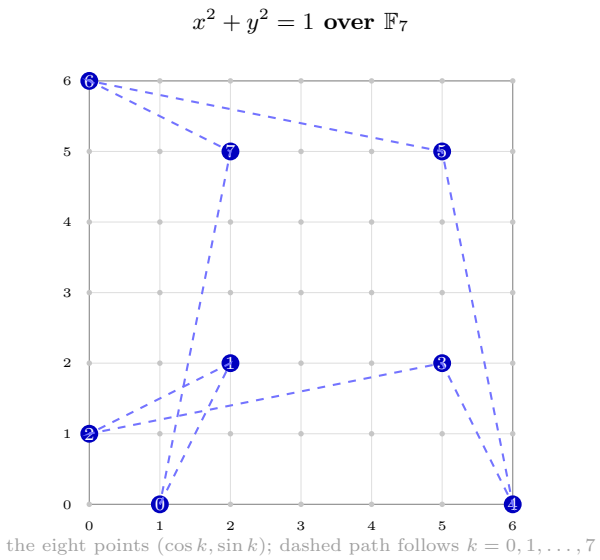


Figure 2: The eight points of $x^2 + y^2 = 1$ over $\mathbb{F}_7$, labelled by the angle $k \in \mathbb{Z}/8$. The dashed path follows $k = 0, 1, \ldots, 7$. This is a conic, not a Euclidean circle: the eight points are scattered in the 7×7 grid.

5.1 Centre, angle group, and order

Does a centre exist? Homogenize the conic as $C : x^2 + y^2 = z^2$ and let $L_\infty : z = 0$. The polar of $[a : b : c]$ is $ax + by - cz = 0$, so the pole of L_∞ is $O = [0 : 0 : 1] = (0, 0)$. The point O is not on C ($0 \neq 1$); it is the unique point with $N(O) = 0$, while every point of C has norm 1. The central symmetry $\sigma(x, y) = (-x, -y)$ preserves C and fixes O; on the angle group it is α^4, the half-turn, and it pairs the eight points into four antipodal pairs $\{k, k + 4\}$ (Figure 3).

The centre is finite exactly when L_∞ is not tangent to C: the ellipse and the hyperbola have a finite centre, while the parabola's centre is the tangency point at infinity.

The angle group around the centre. The symmetries of C fixing O are the orthogonal group $\mathrm{O}_2(\mathbb{F}_7)$ of the quadratic form $x^2 + y^2$, a dihedral group of order 16. Its rotation subgroup is

$$G = \mathrm{SO}_2(\mathbb{F}_7) \cong \mathbb{Z}/8,$$

the norm-one group; it acts regularly on C. The angle of a point P relative to a base point P_0 is the unique $k \in \mathbb{Z}/8$ with $P = \alpha^k P_0$; the rotations are the angle group around the centre.

The full conic automorphism group is $\mathrm{PGL}_2(7)$ of order 336. It does not fix O: the stabilizer of O is $\mathrm{O}_2(\mathbb{F}_7)$ of order 16, so the orbit of O has $336/16 = 21$ points, the internal points of the polarity. Thus the centre is extra structure: the conic alone is $\mathbb{P}^1$ with $\mathrm{PGL}_2(7)$; a centre (equivalently a line at infinity) selects the angle group. This is the same point as the earlier frame discussion: the "circle" is not just the conic.

Order: cyclic, not total. The angle group $\mathbb{Z}/8$ is cyclic; a generator α gives a cyclic order

$$0, 1, 2, \ldots, 7$$

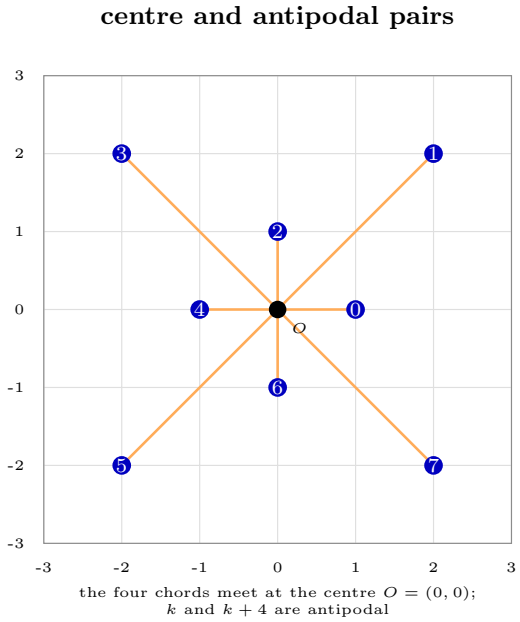


Figure 3: The centre $O = (0, 0)$ is the pole of L_∞. The four chords through O join the antipodal pairs k and $k + 4$. The centre is not on the conic.

on C. The rotations preserve this cyclic order, the reflections in $\mathrm{O}_2(\mathbb{F}_7)$ reverse it, and the full symmetry of the cyclic order is the dihedral group $\mathrm{O}_2(\mathbb{F}_7)$ of order 16.

There is no *total* order invariant under the rotations. An order-preserving permutation of a finite chain is the identity (the least element must be fixed, then inductively all the others), whereas the rotation by one step is an 8-cycle. One can choose a total order by cutting the circle at a base point and declaring $0 < 1 < \cdots < 7$, but the cut depends on the base point and is moved by the rotations; it is not canonical.

This is the finite analogue of the Euclidean circle. The circle has a *cyclic* order (an orientation), not a linear order; the real line has a linear order, the circle does not. The angle k is a coordinate on the cyclic order, not a linear order. The cyclic betweenness is also different from the linear one: for any three points, one lies on the oriented arc from one of the others to the third; this relation is preserved by rotations and reversed by reflections.

model	centre	affine stabiliser	angle group	cyclic order
ellipse	$(0, 0)$	$\mathrm{O}_2(\mathbb{F}_7) \cong D_8$	$\mathbb{Z}/8$	8 points
hyperbola	$(0, 0)$	$\mathrm{O}_{1,1}(\mathbb{F}_7) \cong D_6$	$\mathbb{Z}/6$	6 finite + 2 fixed
parabola	at infinity	D_7	$\mathbb{Z}/7$	7 finite + 1 fixed

Table 4: The centre, the affine stabiliser, and the regular angle group for the three models. The centre is finite iff L_∞ is not tangent; the order of the regular angle group is $q + 1$, $q - 1$, q for the ellipse, hyperbola and parabola.

Table 4 records the three models. In each case the regular angle group is the rotation part of the affine stabiliser of the centre; the points fixed by it are the intersections of L_∞ with C (none

for the ellipse, two for the hyperbola, one for the parabola).

5.2 Arcs, gluing, and transitivity

A directed arc is an interval of the cyclic order. Write $[a, b]_\ell$ for the arc that starts at a and has length $\ell \in \{0, 1, \ldots, 8\}$; its point set is $\{a, a+1, \ldots, a+\ell\}$ modulo 8. The length ℓ is the angle difference $b - a$; $\ell = 0$ is the point a, and $\ell = 8$ is the full circle.

Gluing two arcs is defined when the endpoint of the first is the start of the second:

$$[a, b]_\ell \cdot [b, c]_m = [a, c]_{\ell+m} \qquad (\ell + m \leq 8).$$

If $\ell + m > 8$ the composite wraps more than once and is no longer a simple arc. Thus the simple arcs form a *partial* monoid, not a monoid. If one keeps the winding, i.e. lets the length be an integer in $\mathbb{Z}$ and records the projection $b - a \equiv \ell \pmod 8$, the arcs form the action groupoid of $\mathbb{Z}$ on $\mathbb{Z}/8$: objects are the eight points, arrows are the based arcs, composition is gluing, and the automorphism group of a point is $\mathbb{Z}$ (the full turns). If only nonnegative lengths are allowed, one gets the action category of $\mathbb{N}$; the endomorphism monoid of a point is $\mathbb{N}$ (full turns), not a group.

The question of transitivity has three different answers.

1. *The one-step relation.* For a nonempty set $S \subseteq \mathbb{Z}/8$ of allowed lengths, let $a \, R_S \, b$ iff $b - a \in S$. Then R_S is transitive if and only if S is a subgroup of $\mathbb{Z}/8$. The subgroups are

   $$\{0\}, \qquad \{0, 4\}, \qquad \{0, 2, 4, 6\}, \qquad \mathbb{Z}/8,$$

   with $8, 4, 2, 1$ classes respectively: equality, antipodal pairs, parity, and the universal relation.

2. *Reachability.* The relation R_S^* (a finite sequence of allowed steps) is transitive by construction. Its classes are the cosets of the subgroup $\langle S \rangle$ generated by S; the number of components is

   $$[\mathbb{Z}/8 : \langle S \rangle].$$

   The reachability graph is connected (transitive) exactly when $\langle S \rangle = \mathbb{Z}/8$.

3. *The arc category.* With all lengths allowed, the arc groupoid is transitive (connected): any point reaches any other point. With a restricted set S of lengths, one gets the subgroupoid of $\langle S \rangle$; it is transitive on each coset, not on the whole circle unless S generates $\mathbb{Z}/8$.

Figure 4 shows the first three steps. Table 5 records the components and the one-step transitivity for representative sets.

The figure and the table make the point precise. The full arc structure is transitive, but a restricted arc structure need not be: the arcs of step 2 never connect an even point to an odd one, and the arcs of step 4 only connect antipodal pairs. The components are always the cosets of $\langle S \rangle$.

This is the finite analogue of the fundamental groupoid of the circle. The topological circle has path groupoid with automorphism group $\pi_1(S^1) = \mathbb{Z}$; the finite arc groupoid with winding has the same automorphism group $\mathbb{Z}$, and its projection to the eight points is the quotient by the full turn 8. The simple arcs are the fundamental category; the full turns are the endomorphisms.

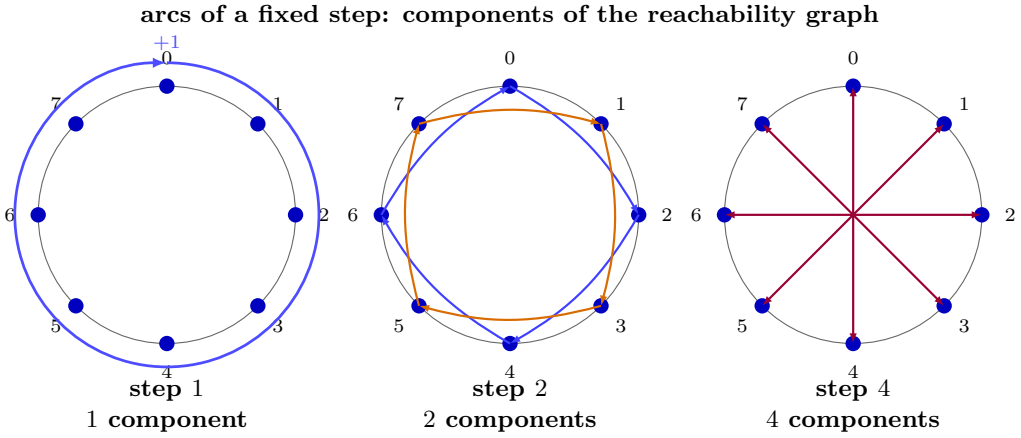


Figure 4: Arcs of a fixed step k: the components of the reachability graph. $k = 1$ gives one 8-cycle; $k = 2$ gives two 4-cycles (parity); $k = 4$ gives four 2-cycles (antipodal pairs).

S	$\langle S \rangle$	components	R_S transitive?
$\{1\}$	$\mathbb{Z}/8$	1	no
$\{2\}$	$\{0, 2, 4, 6\}$	2 (parity)	no
$\{3\}$	$\mathbb{Z}/8$	1	no
$\{4\}$	$\{0, 4\}$	4 (antipodal)	no
$\{2, 4\}$	$\{0, 2, 4, 6\}$	2	no
$\{0, 4\}$	$\{0, 4\}$	4	yes
$\mathbb{Z}/8$	$\mathbb{Z}/8$	1	yes

Table 5: Allowed arc lengths S, the subgroup they generate, the number of reachability components, and whether the one-step relation is transitive. For nonempty S the one-step relation is transitive exactly for the subgroups; reachability is transitive exactly when S generates $\mathbb{Z}/8$.

5.3 Experiment: translating the circle

Translate every point of C by one cell upward:

$$T(x, y) = (x, y + 1).$$

The image of the conic is

$$C' : \quad x^2 + (y - 1)^2 = 1,$$

with the eight points

$$(1, 1),\ (2, 3),\ (0, 2),\ (5, 3),\ (6, 1),\ (5, 6),\ (0, 0),\ (2, 6).$$

The unit-circle relation is *not* preserved: none of the new points has norm 1 with respect to the original norm. Their norms are

$$2,\ 6,\ 4,\ 6,\ 2,\ 5,\ 0,\ 5,$$

so the new set is not the norm-one fibre of $N(x, y) = x^2 + y^2$. The centre moves from $O = (0, 0)$ to $O' = T(O) = (0, 1)$. The two centres exchange roles: the old centre O lies on C' because $0 + (0 - 1)^2 = 1$, and the new centre O' lies on C because $0 + 1^2 = 1$. The two conics are disjoint: $C \cap C'$ would require

$$y^2 = (y - 1)^2 \implies 2y = 1 \implies y = 4, \qquad x^2 + 16 = x^2 + 2 = 1 \implies x^2 = 6,$$

and 6 is a nonsquare in $\mathbb{F}_7$.

The image is nevertheless *still a conic*. The reason is that T is an affine transformation, and affine transformations preserve conics. In fact T fixes L_∞ pointwise, so C' is again an ellipse; the angle group is the conjugate

$$T\,G\,T^{-1} \cong \mathbb{Z}/8,$$

which fixes O' and acts regularly on C'. The cyclic order and the arc structure are transported by T; transitivity is unchanged.

	original C	translated C'
equation	$x^2 + y^2 = 1$	$x^2 + (y - 1)^2 = 1$
centre	$(0, 0)$	$(0, 1)$
norm $N(x, y) = x^2 + y^2$	constant 1	$0, 2, 4, 5, 6$
L_∞	disjoint	disjoint
angle group	G	$TGT^{-1} \cong \mathbb{Z}/8$

Table 6: The unit circle and its translate. The conic property and the projective/affine structure survive; the centre and the norm-one relation do not.

The moral is the distinction between the projective and the affine data. A conic is preserved by every projectivity, in particular by every translation; the unit-circle data — the centre O, the norm N, and the angle group around O — are not. A uniform translation gives another conic (a circle with a new centre), not the same circle. If the eight points were moved by an arbitrary permutation instead of a translation, the image would generally not be a conic at all; the conic property is projective/affine-invariant, not permutation-invariant.

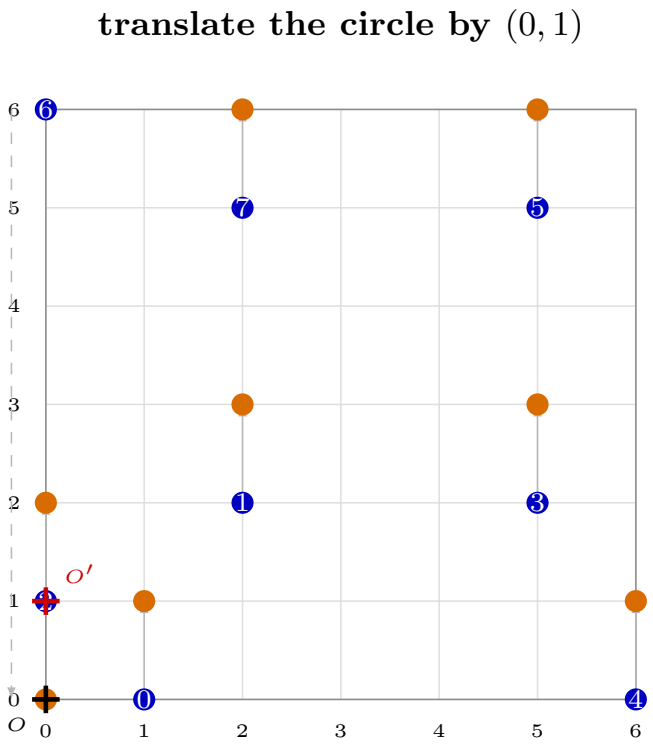


Figure 5: The original conic (blue, $x^2 + y^2 = 1$) and its translate (orange, $x^2 + (y-1)^2 = 1$) under $(x, y) \mapsto (x, y+1)$. The two conics are disjoint; the old centre $O = (0, 0)$ is a point of the orange conic, and the new centre $O' = (0, 1)$ is a point of the blue one.

5.4 Movable circles and transportability

The right definition is not a fixed subset but a relative relation. For a centre O and a radius parameter $r \in \mathbb{F}_7$, put

$$S(O, r) = \{P : Q(P - O) = r\}, \qquad Q(v) = v_1^2 + v_2^2.$$

The unit circle is $S(0, 1)$; the translated circle of the previous experiment is $S((0, 1), 1)$. The relation is translation-equivariant:

$$T\big(S(O, r)\big) = S\big(T(O), r\big)$$

for every translation T. The absolute norm $N(P) = Q(P)$ is not translation-invariant; the relative norm $Q(P - O)$ is. That is the movable circle relation. In Euclidean language r is the square of the radius; over $\mathbb{F}_7$ the square root need not exist, so r is the honest parameter.

The family of circles is

$$\mathcal{C} = \{(O, r) : O \in \mathbb{F}_7^2,\ r \in \mathbb{F}_7\}.$$

The translations act transitively on the centres with r fixed; the similarity group (translations, orthogonal transformations, dilations) acts on the radius by $r \mapsto \lambda^2 r$. The group actions are summarized in Table 7.

group	action on (O, r)	preserves circles?
translations	$O \mapsto O + t$, r fixed	yes
similarities	$O \mapsto \lambda RO + t$, $r \mapsto \lambda^2 r$	yes
affine	$O \mapsto AO + t$	only if $A = \lambda R$; otherwise circles $\to$ ellipses
projective	general	only if $\{I, J\}$ is preserved; otherwise circles $\to$ conics

Table 7: The circle is a similarity structure, not an affine or projective one. I, J are the circular points at infinity.

The circular points make the distinction precise. Homogenize $Q(x, y) = x^2 + y^2$ and put $z = 0$: the two points

$$I = [1 : \mathrm{i} : 0], \qquad J = [1 : -\mathrm{i} : 0]$$

in $\mathbb{P}^2(\mathbb{F}_{49})$ satisfy $x^2 + y^2 = 0$. A conic is a circle exactly when it passes through I and J. The similarity group is the stabilizer of the pair $\{I, J\}$; a general affine map acts on the line at infinity by its linear part and moves I, J unless the linear part is a similarity, so it sends a circle to an ellipse; a general projectivity sends a circle to a conic not through I, J. The conic is projective; the circle is a similarity notion.

The angle group is transported with the centre. The stabilizer of O in the similarity group is $\mathrm{O}_2(\mathbb{F}_7)$; its rotation subgroup is $G = \mathrm{SO}_2(\mathbb{F}_7) \cong \mathbb{Z}/8$. For the translated circle the angle group is TGT^{-1}, and the cyclic order and the arc structure are transported by T. Thus the whole structure — centre, circle, angle group, cyclic order, arcs — is transportable under translations, and under similarities up to the square class of r.

The square classes are the only obstruction to moving between different radii:

$$\mathbb{F}_7^\times / \mathbb{F}_7^{\times 2} = \{\{1, 2, 4\}, \{3, 5, 6\}\}.$$

The unit circle is similar to the circles of square radius, not to those of nonsquare radius. Over $\mathbb{R}$ the positive radii form one orbit; over $\mathbb{C}$ all nonzero radii form one orbit. This is the same square-class phenomenon as the earlier trichotomy.

5.5 Scaling and the ripple pencil

There are two different families of circles attached to the centre 0, and the scaling group acts on them differently.

The norm circles. The level sets

$$S_r = \{P : Q(P) = r\}, \qquad r \in \mathbb{F}_7^\times,$$

are the six 8-point conics centred at 0; $S_0 = \{0\}$. The dilation $D_\lambda(P) = \lambda P$ acts by

$$D_\lambda(S_r) = S_{\lambda^2 r}.$$

Since λ^2 runs over the squares $\{1, 2, 4\}$, the six circles split into two scaling orbits,

$$\{S_1, S_2, S_4\}, \qquad \{S_3, S_5, S_6\},$$

the square classes of the radius.

The ripple pencil. The clue horocycles in the finite model are the horizontal lines

$$H_t = \{y = t\}, \qquad t \in \mathbb{F}_7,$$

based at the point at infinity ∞. The inversion

$$J(z) = -\frac{1}{z}$$

maps H_t to the conic

$$C_t = J(H_t): \quad x^2 + y^2 = \frac{y}{t}, \qquad t \in \mathbb{F}_7^\times.$$

Every C_t is an 8-point conic through the pole 0 (the point ∞ maps to 0); the six of them form the finite *ripple pencil*. The correspondence

$$J(H_t) = C_t$$

is exact: the point path (the clue) and the ripple history are two readings of the same Möbius map, as in the path–ripple duality of Paper 0.

Under the same dilation, the ripple parameter transforms by

$$D_\lambda(C_t) = C_{t/\lambda}.$$

Here λ runs over all of $\mathbb{F}_7^\times$, not just the squares, so the six ripple conics form a *single* orbit

$$\{C_1, C_2, \ldots, C_6\}.$$

This is the precise finite form of the nested family of Euclidean circles tangent at 0: the nesting parameter t is the height of the clue line, and the scaling acts on it by the inverse scalar λ^{-1}.

The visual point is worth stating plainly. The Euclidean picture of the ripple family is a nested family of circles tangent at 0; the nesting is an ordered notion (inside/outside, radius). Over $\mathbb{F}_7$ there is no such order: the six C_t are 8-point conics through 0, and their arrangement on the 7×7 grid is a coloured partition, not a sequence of nested rings. What is observable is the *pencil* (all six through the pole), the *partition* of the grid, and the scaling action; what is not observable is the Euclidean nesting. The correspondence $J(H_t) = C_t$ and the scaling laws $t \mapsto \lambda^{-1}t$, $r \mapsto \lambda^2 r$ hold exactly; they are algebraic statements, not visual ones.

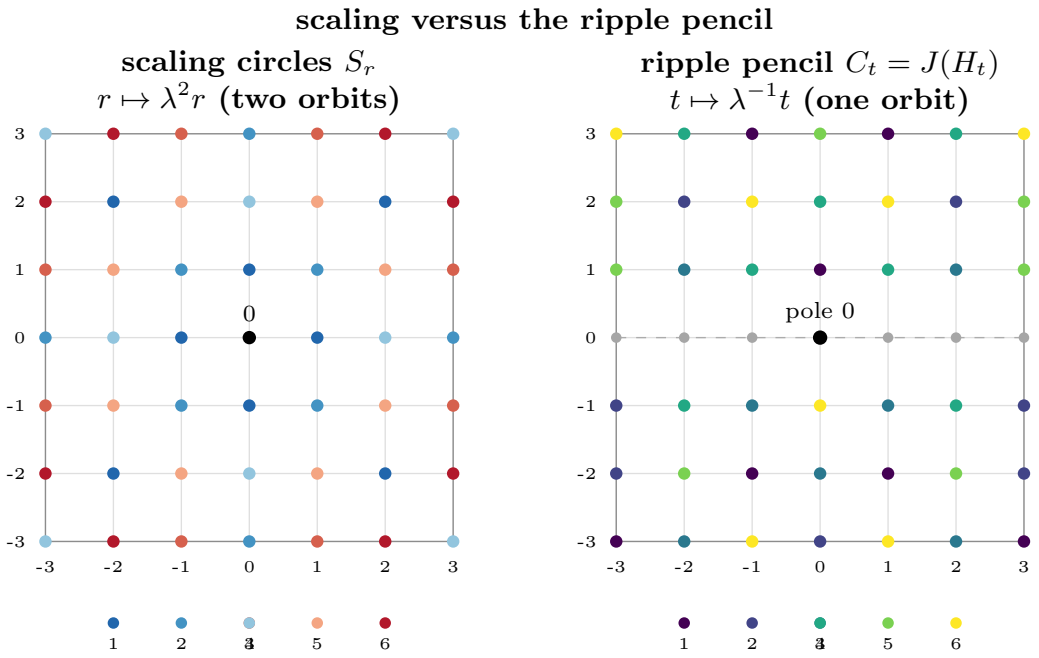


Figure 6: Left: the norm circles S_r (level sets of Q), coloured by r; their scaling orbits are the two square classes. Right: the ripple conics $C_t = J(H_t)$, all through the pole 0; the grey points on the x-axis are the clue line H_0. In the Euclidean upper half-plane the C_t are nested circles tangent at 0; over $\mathbb{F}_7$ they form a pencil of six 8-point conics through 0.

family	equation	common point	scaling	orbits
clue lines H_t	$y = t$	∞	$t \mapsto \lambda t$	one
ripple conics C_t	$x^2 + y^2 = y/t$	0	$t \mapsto \lambda^{-1} t$	one
norm circles S_r	$x^2 + y^2 = r$	none	$r \mapsto \lambda^2 r$	two square classes

Table 8: The clue lines H_t and the ripple conics $C_t = J(H_t)$ are exchanged by the inversion $J(z) = -1/z$; scaling acts linearly on t. The norm circles S_r are a different family, with the square-class obstruction.

5.6 The real picture and the finite picture

It is worth placing the finite ripple pencil next to the real one from Paper 0. Over $\mathbb{R}$, with $J(z) = -1/z$ and $H_t = \{y = t\}$, the image is the circle

$$C_t : \quad x^2 + \left(y - \frac{1}{2t}\right)^2 = \left(\frac{1}{2t}\right)^2, \qquad t > 0.$$

All these circles are tangent to the real axis at 0 and nested as t varies; the parameter t is continuous. The clue lines H_t and the ripple circles C_t are exchanged by J:

$$J(H_t) = C_t.$$

The dilation D_λ, $\lambda > 0$, sends t to t/λ; the family is a single orbit, and the nesting is visible.

Over $\mathbb{F}_7$ the same equations produce the six conics

$$C_t : \quad x^2 + y^2 = \frac{y}{t}, \qquad t \in \mathbb{F}_7^\times.$$

They all pass through the pole 0, and they form a *pencil*, not a nested sequence. The correspondence $J(H_t) = C_t$ and the scaling law $t \mapsto t/\lambda$ still hold; the six form one orbit. What is lost is the order: there is no inside/outside, no continuous parameter, no nesting.

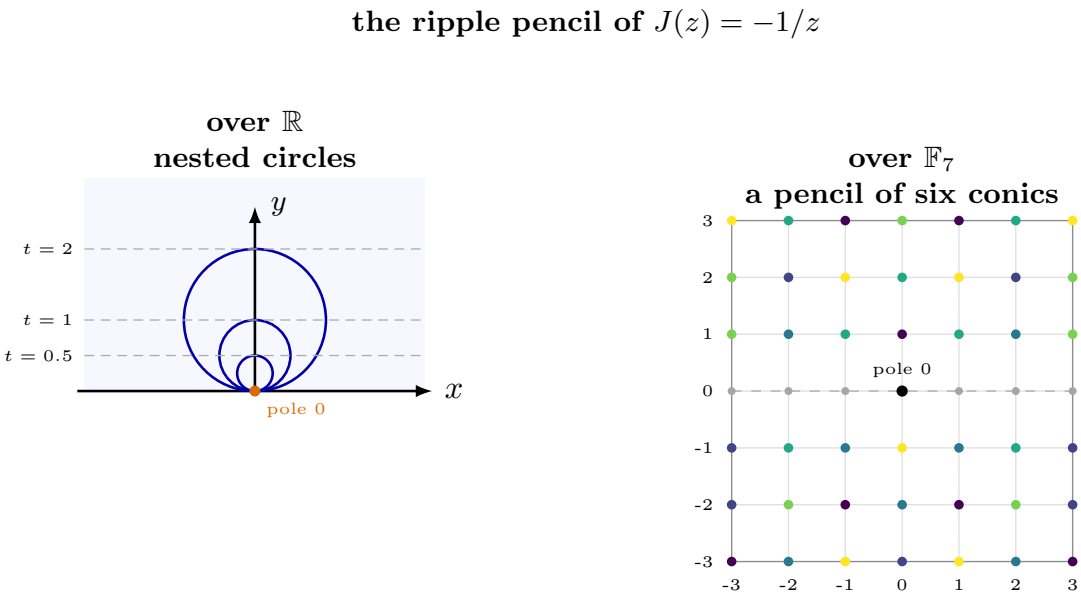


Figure 7: The ripple pencil of $J(z) = -1/z$ over $\mathbb{R}$ and over $\mathbb{F}_7$. Left: the clue lines H_t (dashed) map to nested circles tangent at 0. Right: the same equations over $\mathbb{F}_7$ give six 8-point conics through 0; the family is a pencil, not a nested sequence.

The moral is the same one that has recurred throughout these notes: the *algebraic* structure — the conic family, the Möbius correspondence, the scaling orbit — is field-independent, while the *order* structure — nesting, inside/outside, continuous parameter — is special to $\mathbb{R}$. The finite "ripple geometry" is a pencil, not a new geometry; the name is descriptive, as Paper 0 says of its real counterpart.

	over $\mathbb{R}$	over $\mathbb{F}_7$
parameter	$t > 0$ continuous	$t \in \mathbb{F}_7^\times$ (six values)
C_t	nested circles tangent at 0	8-point conics through 0
ordering	nesting by t	no order; pencil
scaling	$t \mapsto t/\lambda$, $\lambda > 0$	$t \mapsto t/\lambda$, $\lambda \in \mathbb{F}_7^\times$
orbits	one	one
$J(H_t) = C_t$	yes	yes
visual	nested rings	coloured pencil

Table 9: The real and finite ripple pencils. The algebraic correspondence and the scaling orbit are field-independent; the nesting is a real/order phenomenon.

6 The angle group

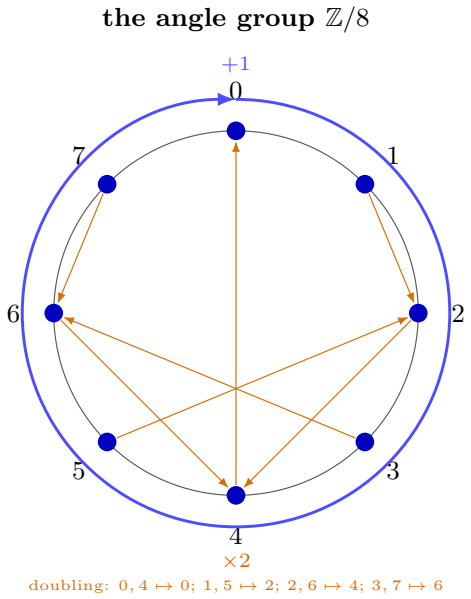


Figure 8: The angle group $\mathbb{Z}/8$. The blue outer arrow is the $+1$ cycle; the orange chords are the doubling map $k \mapsto 2k$. Doubling is two-to-one: $0, 4 \mapsto 0$; $1, 5 \mapsto 2$; $2, 6 \mapsto 4$; $3, 7 \mapsto 6$.

The angle group is the cyclic group $\mathbb{Z}/8$. Its automorphism group is $(\mathbb{Z}/8)^\times = \{1, 3, 5, 7\}$; multiplication by $3, 5, 7$ permutes the eight angles, while multiplication by 2 collapses pairs. The addition formula of Proposition 4.1 is the statement that the conic's group law and the angle group's addition are the same operation.

7 Euclidean angle axioms and the finite angle group

The phrase "angle domain" has two readings. In Euclidean geometry it usually means the structure of angles: congruence, addition, order, and measure. It can also mean an angular region (a sector) between two rays. The finite model has a clean analogue of the first reading and only a combinatorial analogue of the second; this section compares the first.

7.1 The Euclidean angle structure

Three groups of axioms are relevant.

1. *Congruence.* Angle congruence is an equivalence relation. Given an angle and a ray, there is a unique angle on a prescribed side of the ray congruent to the given one (Hilbert's copying axiom); the SAS criterion and the rest of Euclidean congruence follow.

2. *Protractor.* Angles have a real-valued measure $\mu(\angle AOB) \in [0, \pi]$ (or in $\mathbb{R}/2\pi\mathbb{Z}$ for oriented angles), additive when a ray lies inside the angle. The measure is invariant under isometries.

3. *Order and continuity.* Rays from a point in a half-plane are totally ordered by betweenness, so unoriented angle measures in $[0, \pi]$ can be compared. The Archimedean (continuity) axiom makes the measure real-valued and allows an angle to be divided into n equal parts for every n.

The standard theorems are then: supplements and complements are unique; a right angle is congruent to its supplement; vertical angles are equal; every angle has a unique bisector; the angles of a triangle sum to π (equivalent to the parallel postulate); the exterior-angle theorem; the isosceles base-angle theorem; the inscribed-angle theorem; Thales' theorem; and the AA criterion for similarity.

In group language, oriented angles form $\mathbb{R}/2\pi\mathbb{Z}$; the rotation group $\mathrm{SO}(2)$ acts regularly on the circle, and the angle from a base ray to a target ray is the group element taking one to the other.

7.2 The finite analogue

On the conic $C \cong \mathbb{P}^1(\mathbb{F}_7)$, the norm-one group $G = \mathbb{Z}/8$ acts regularly: for $P, Q \in C$ there is a unique $h \in \mathbb{Z}/8$ with $Q = \alpha^h P$. This h is the finite angle from P to Q, and the map $h \mapsto \alpha^h$ is the finite protractor. Addition of angles is addition in $\mathbb{Z}/8$; the right angle is 2 (because $4 \cdot 2 = 8 = 0$), the straight angle is 4, and the full angle is 0. Figure 9 shows the two protractors side by side.

Table 10 records which Euclidean properties survive.

7.3 Three concrete comparisons

Example 7.1 (Bisection)**.** In Euclidean geometry every angle $0 < \alpha < \pi$ has a unique bisector. In $\mathbb{Z}/8$ the bisector equation is $2h = k$. If k is even there are two solutions h and $h + 4$, the two orientations of the same bisector line; the finite group has no order with which to choose one. If k is odd there is no solution. For example, the right angle $k = 2$ has $h = 1, 5$, while $k = 1$ has no bisector. This is exactly the failure of divisibility by 2.

Proposition 7.2 (Finite Thales)**.** *Let $A, B \in C$ be antipodal, so $B = -A$, and let $P \in C$. Then*

$$(A - P) \cdot (B - P) = A \cdot B - A \cdot P - P \cdot B + P \cdot P = -1 - A \cdot P + A \cdot P + 1 = 0.$$

Thus the vectors PA and PB are orthogonal with respect to the standard dot product on $\mathbb{F}_7^2$: the angle in a semicircle is right, in the bilinear sense.

Proof. $A \cdot B = -1$ because $B = -A$ and $A \cdot A = 1$; $P \cdot P = 1$ because $P \in C$; and $P \cdot B = P \cdot (-A) = -A \cdot P$. The sum is zero. □

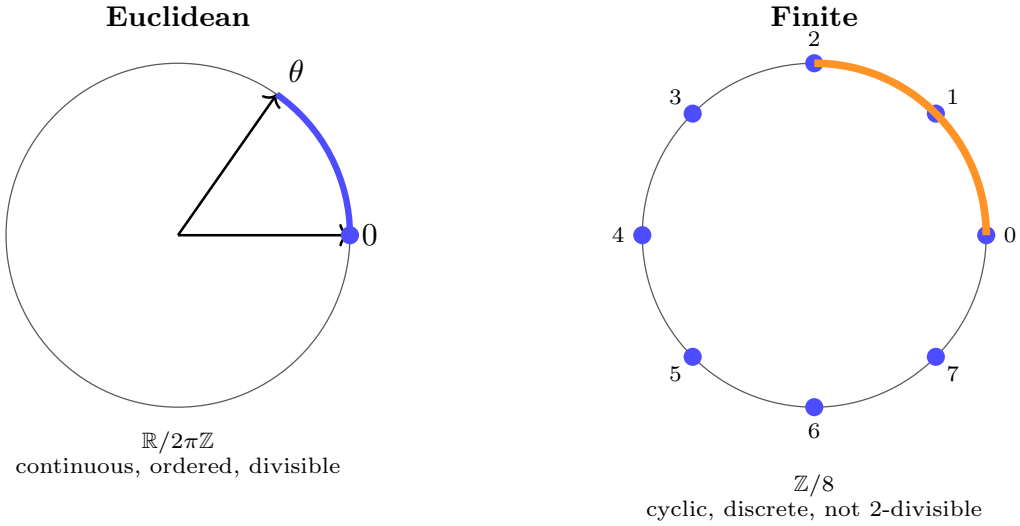


Figure 9: The continuous protractor $\mathbb{R}/2\pi\mathbb{Z}$ and the finite protractor $\mathbb{Z}/8$. On the right the eight points of C are the eight angles; the orange arc $0 \to 2$ is the right angle, passing through the angle 1.

Euclidean	finite $\mathbb{Z}/8$	verdict
angle group $\mathbb{R}/2\pi\mathbb{Z}$	$\mathbb{Z}/8$ on C	different group
protractor additivity	$k + l \bmod 8$	holds
right angle $\pi/2$; four right $= 2\pi$	2; $4 \cdot 2 = 8 = 0$	holds
supplement $\alpha + \beta = \pi$, unique	$k + l = 4$; $l = 4 - k$	holds
complement $\alpha + \beta = \pi/2$, unique	$k + l = 2$; $l = 2 - k$	holds
congruence is an equivalence	equality in $\mathbb{Z}/8$	holds
total order and betweenness	cyclic order only	fails
angle copying	translation $k \mapsto k + h$	holds
unique bisector	$2h = k$: 0 or 2 solutions	fails in general
n-section, all n	$nh = k$ iff $\gcd(n, 8) \mid k$	fails in general
vertical angles equal	no angle between lines	not defined
triangle angle sum π	no triangle angle measure	not defined
inscribed angle = half arc	2 not invertible	only 2-to-1
Thales (semicircle)	$(A - P) \cdot (B - P) = 0$	holds
similarity and AA	no metric or angle	fails
continuous measure	$\mathbb{Z}/8$-valued measure	fails

Table 10: Euclidean angle axioms and theorems versus the finite angle group $\mathbb{Z}/8$. The first row records the same algebraic shape (a regular abelian group action) with a different group. "Holds" means the algebraic statement survives; "fails" means the Euclidean proof uses order, continuity, or division, which $\mathbb{Z}/8$ lacks.

Remark 7.3 (Triangle angle sum and inscribed angle). The Euclidean triangle angle sum is π, and the inscribed angle is half the subtended arc. On C the arc between two points is an element of $\mathbb{Z}/8$; halving it requires solving $2h = \text{arc}$. When the arc is odd there is no solution, and when it is even there are two. Thus the inscribed-angle theorem has only a 2-to-1 shadow in $\mathbb{Z}/8$. A triangle in the finite affine plane has no Euclidean angle measure at its vertices: the angle group $\mathbb{Z}/8$ is attached to the conic, not to arbitrary lines. The triangle angle sum is therefore not a theorem of this finite model; its Euclidean proof uses the order and the parallel postulate, neither of which is present.

7.4 What the comparison says

The common algebraic skeleton is the angle group: an abelian group acting regularly on the circle or conic, with the angle as the group element. Euclidean geometry adds three things that $\mathbb{Z}/8$ lacks: a total order compatible with addition, an Archimedean continuous measure, and divisibility (the ability to solve $nh = k$ for every n).

The theorems that depend only on the group law survive: addition, supplements, complements, right and straight angles, copying, and finite Thales. The theorems that use order, continuity, or division fail, or have only a 2-to-1 shadow: bisection, n-section, the triangle angle sum, vertical angles, and similarity. This is the same pattern as the $\mathbb{R}/\mathbb{F}_7$ trichotomy: the algebraic classification survives; the continuous invariants do not.

8 Stereographic and half-angle coordinates

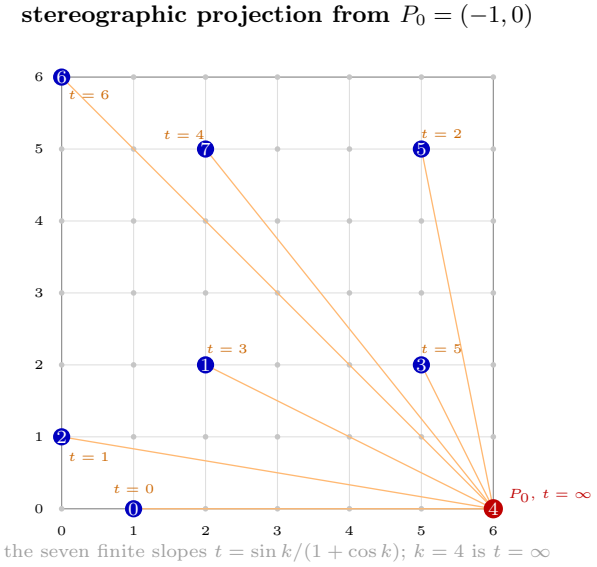


Figure 10: Stereographic projection from $P_0 = (-1, 0) = (6, 0)$. The seven finite slopes $t = \sin k/(1 + \cos k)$ give the seven points other than P_0; the point P_0 itself is $t = \infty$. The Euclidean segments are schematic: a projective line over $\mathbb{F}_7$ contains eight points and wraps around the grid.

For $k \neq 4$ (so $\alpha^k \neq -1$), the line through $P_0 = (-1, 0)$ and $(\cos k, \sin k)$ has slope

$$t = \frac{\sin k}{1 + \cos k} \in \mathbb{F}_7. \tag{1}$$

The inverse formulas are

$$\cos k = \frac{1 - t^2}{1 + t^2}, \qquad \sin k = \frac{2t}{1 + t^2}, \tag{2}$$

and $k = 4$ corresponds to $t = \infty$. Thus

$$k \longmapsto t$$

is a bijection $\mathbb{Z}/8 \setminus \{4\} \to \mathbb{F}_7$, i.e. a finite half-angle (tangent) coordinate. The formulas (1)–(2) are verified for all k in `verify.py`.

9 Discrete waveforms and the first difference

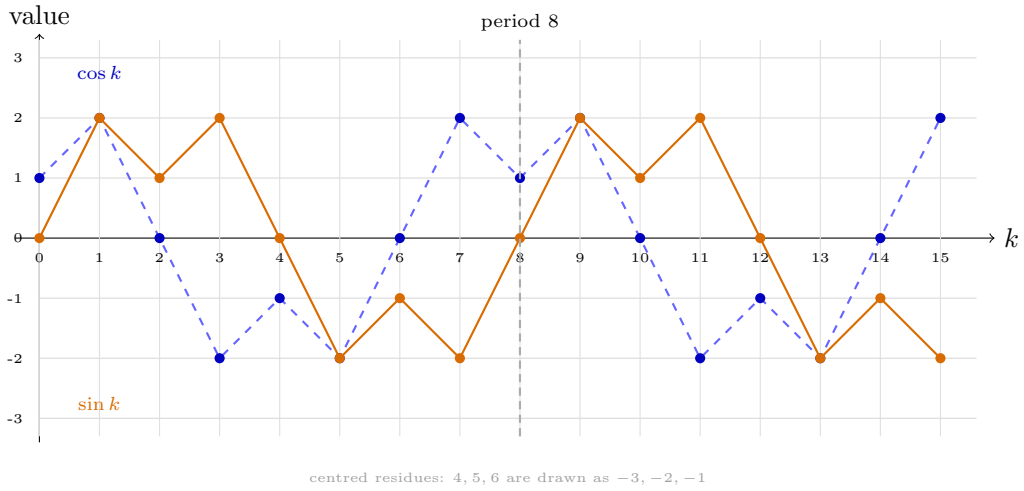


Figure 11: $\cos k$ and $\sin k$ as functions on $\mathbb{Z}/8$, repeated over two periods. The values are centred residues in $\{-3, \ldots, 3\}$; the dashed vertical line marks the period 8. These are discrete waveforms, not continuous graphs.

The functions cos and sin are periodic functions on the finite group $\mathbb{Z}/8$. Their first finite difference is

$$\Delta f(k) = f(k + 1) - f(k).$$

For example, with centred representatives,

$$\Delta \cos = (1, -2, -2, 1, -1, 2, 2, -1), \qquad \Delta \sin = (2, -1, 1, -2, -2, 1, -1, 2),$$

and in both cases the sum over the cycle is 0:

$$\sum_{k=0}^{7} \Delta f(k) = f(8) - f(0) = 0.$$

This is the discrete fundamental theorem of calculus: summation and difference are inverse up to the value at 0. There is no limit, no derivative as a limit, and no mean value theorem; the calculus here is finite and algebraic. A systematic treatment of Δ, summation, Hasse derivatives and residues on $\mathbb{P}^1(\mathbb{F}_7)$ is the next note.

10 The harmonic layer: characters of $\mathbb{Z}/8$

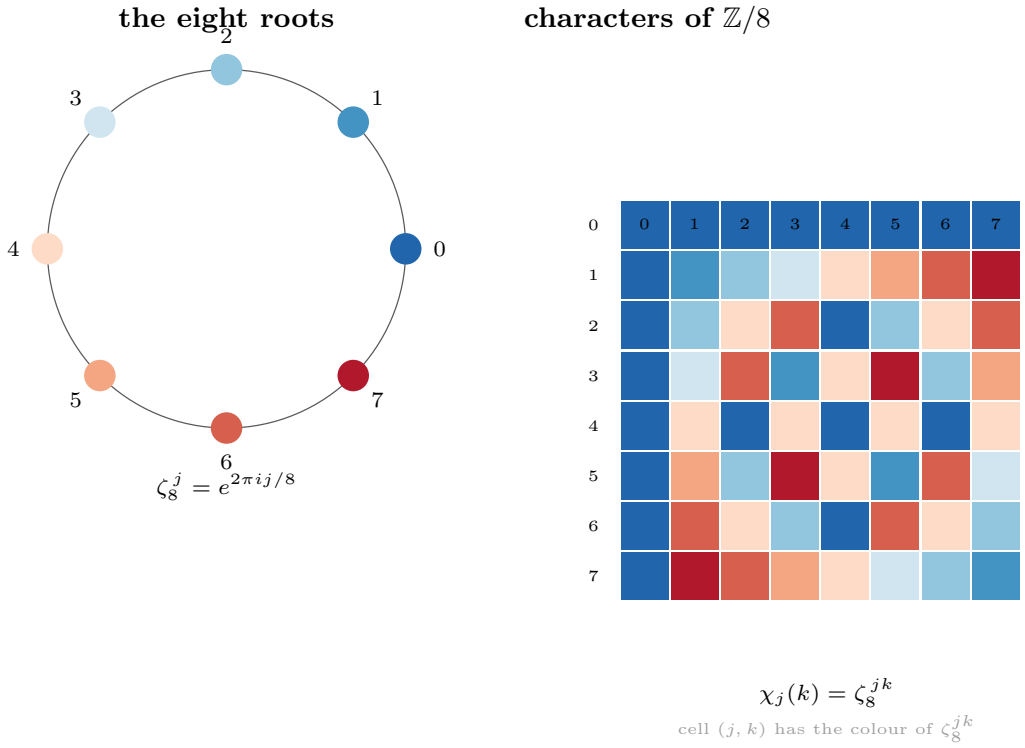


Figure 12: The harmonic layer. Left: the eight eighth roots of unity ζ_8^j. Right: the character matrix $\chi_j(k) = \zeta_8^{jk}$; the cell (j,k) has the colour of ζ_8^{jk}. The colours are a legend shared by the two panels.

The algebraic trigonometry of the previous sections takes values in $\mathbb{F}_7$. A second, harmonic layer takes values in $\mathbb{C}$. Let

$$\zeta_8 = e^{2\pi \mathrm{i}/8}$$

and define the characters of $\mathbb{Z}/8$ by

$$\chi_j(k) = \zeta_8^{jk}, \qquad j,k \in \mathbb{Z}/8.$$

For a function $f : \mathbb{Z}/8 \to \mathbb{C}$, the finite Fourier transform is

$$\widehat{f}(j) = \sum_{k=0}^{7} f(k)\, \zeta_8^{-jk},$$

with inversion

$$f(k) = \frac{1}{8} \sum_{j=0}^{7} \widehat{f}(j)\, \zeta_8^{jk}$$

and Parseval

$$\sum_{k=0}^{7} |f(k)|^2 = \frac{1}{8} \sum_{j=0}^{7} |\widehat{f}(j)|^2.$$

The transform diagonalizes translations: if $(\tau_a f)(k) = f(k-a)$, then $\widehat{\tau_a f}(j) = \zeta_8^{-ja} \widehat{f}(j)$. For the general theory of Fourier analysis on finite groups see [2].

It is important not to identify the two layers. The functions $\cos k, \sin k \in \mathbb{F}_7$ are the coordinates of α^k on the conic; the characters $\chi_j(k) \in \mathbb{C}$ are the Fourier basis of the angle group. They share the index group $\mathbb{Z}/8$, but they take values in different fields and answer different questions. The bridge between the additive characters of $\mathbb{F}_7$ and the multiplicative characters of $\mathbb{F}_7^\times$ is the Gauss sum

$$G(\chi) = \sum_{x \in \mathbb{F}_7} \chi(x)\, \zeta_7^x,$$

which is the finite analogue of the Gamma integral; it will be treated in the next note.

11 What is not claimed, and what comes next

This note is deliberately small. It does *not* claim:

- a real or complex analysis over $\mathbb{F}_7$: there are no nontrivial limits, no continuity in the analytic sense, no mean value theorem;
- that the finite trigonometry is a picture of the real one: the values are in $\mathbb{F}_7$, the angle group is $\mathbb{Z}/8$, and the identities are algebraic;
- that the conic or Pascal's theorem alone gives analysis: the conic gives the geometry and the group law; the analysis comes from the derivation, difference, representation and cohomological structures built on top.

The natural next steps are:

1. discrete calculus on $\mathbb{P}^1(\mathbb{F}_7)$: Δ, summation, Hasse derivatives, differential forms, residues and the residue theorem;
2. Gauss sums and the bridge between the additive and multiplicative characters;
3. the harmonic analysis of functions on $\mathrm{PGL}_2(\mathbb{F}_7)$ and the Bruhat–Tits tree [1], as a non-Archimedean analogue of the hyperbolic analysis of Paper 0/Paper I;
4. the characteristic-7 Magnus expansion and the group algebra $\mathbb{F}_7[\mathrm{PGL}_2(\mathbb{F}_7)]$, as the finite analogue of the series pillar of Paper I.

References

[1] J.-P. Serre, *Trees*, Springer, 1980.

[2] A. Terras, *Fourier Analysis on Finite Groups and Applications*, Cambridge University Press, 1999.

[3] R. Lidl and H. Niederreiter, *Finite Fields*, Cambridge University Press, 1997.